

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-4

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

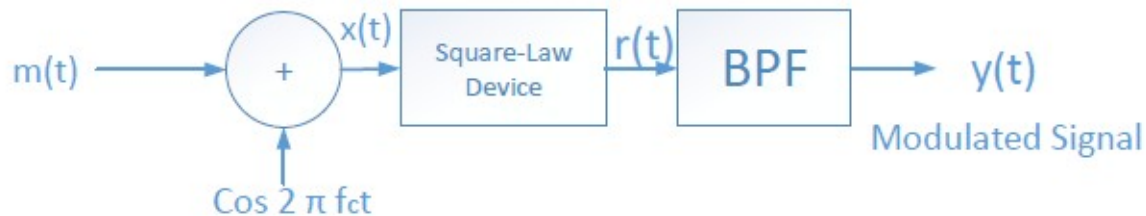
Date: 03-09-2025

Objectives

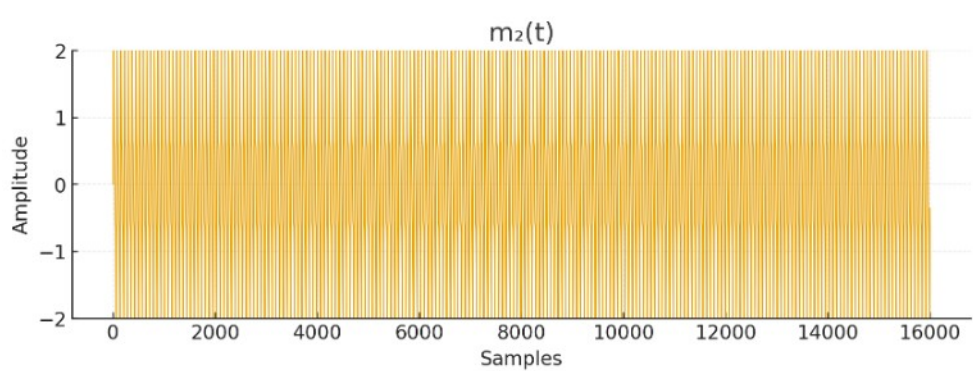
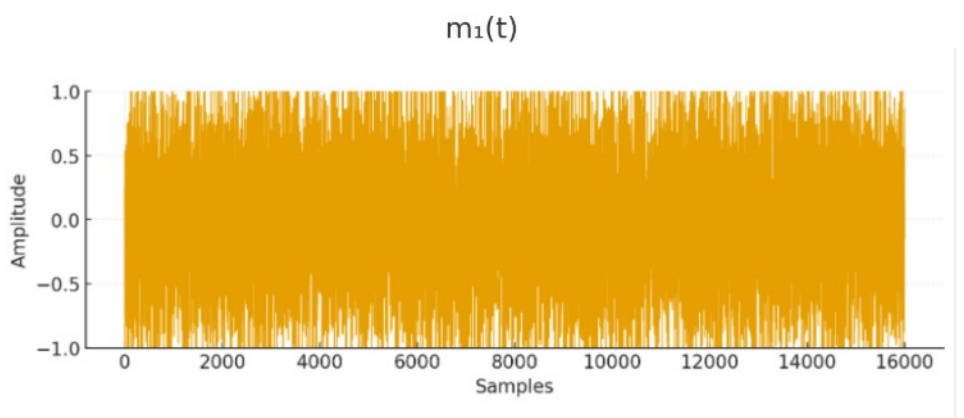
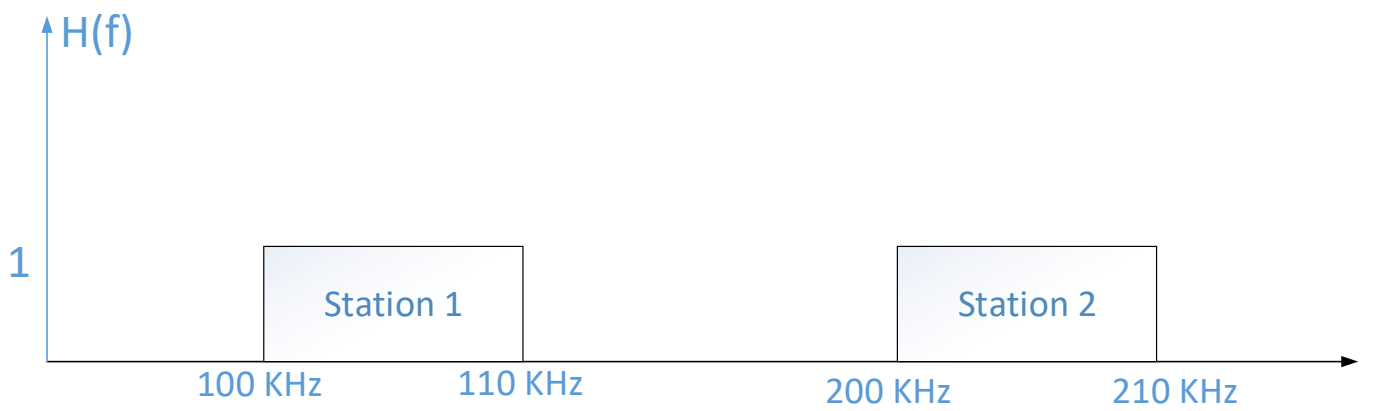
- Modulation, DSB-SC, AM

Tutorial Problems

1. Find the power efficiency for an AM signal $x(t) = \cos 220\pi t + 4 \cos 200\pi t + \cos 180\pi t$.
2. An analog passband transmission is shown in the figure below. The bandpass filter has a bandwidth $2B$ centered at f_c . The message signal $m(t)$ is a real-valued signal with amplitude varying between $-A_m$ to A_m , and has an average power P_m . The bandwidth of $m(t)$ is B . Assume $f_c \gg B$. The square-law device obeys $r(t) = 2x(t) + x^2(t)$.



- (a) Find the average transmitted power of the modulated signal $y(t)$ in terms of P_m .
 - (b) Find the modulation index of the modulated signal $y(t)$.
 - (c) Draw a system block diagram for coherent demodulation.
 - (d) What is range value of A_m that will ensure non-coherent demodulation?
3. There are two radio stations Station 1 and Station 2 transmitting audio messages $m_1(t)$ and $m_2(t)$, respectively. Available channel bandwidth and messages are shown in figures. Design both the transmitter and receiver using Amplitude Modulation (AM) with envelope detection, assuming a modulation index of 1.



Solution

1. Using

$$\cos(220\pi t) + \cos(180\pi t) = 2 \cos(200\pi t) \cos(20\pi t).$$

$$x(t) = 4 \cos(200\pi t) + 2 \cos(200\pi t) \cos(20\pi t).$$

$$x(t) = [4 + 2 \cos(20\pi t)] \cos(200\pi t).$$

Standard AM form,

$$x(t) = [A + m(t)] \cos(2\pi f_c t),$$

$$m(t) = 2 \cos(20\pi t), \quad c(t) = 4 \cos(200\pi t).$$

Carrier power $16/2 = 8$; message power: $P_m = 4/2 = 2$; Sideband signal $m(t) \cos(200\pi t)$; Sideband power $P_m/2 = 1$; Power efficiency $\eta = P_{SB}/P_{total} = 1/(1 + 8) = 1/9 = 11.1\%$

2. Signal $x(t) = m(t) + \cos 2\pi f_c t$
 $r(t) = 2x(t) + x^2(t)$

$$r(t) = 2m(t) + 2 \cos(2\pi f_c t) + m^2(t) + 2m(t) \cos(2\pi f_c t) + \cos^2(2\pi f_c t).$$

Use the trigonometric identity

$$\cos^2(2\pi f_c t) = \frac{1}{2}(1 + \cos(4\pi f_c t)),$$

$$r(t) = 2m(t) + 2 \cos(2\pi f_c t) + m^2(t) + 2m(t) \cos(2\pi f_c t) + \frac{1}{2} + \frac{1}{2} \cos(4\pi f_c t).$$

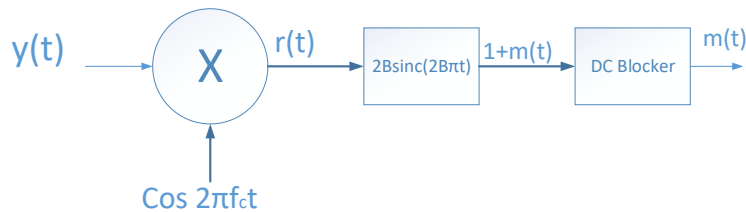
BPF at f_c will pass the signals: $2 \cos(2\pi f_c t)$ and $2m(t) \cos(2\pi f_c t)$ and suppress other terms.
 $y(t) = 2 \cos(2\pi f_c t) + 2m(t) \cos(2\pi f_c t).$

(a) $P_y = \frac{4}{2} + \frac{4P_m}{2} = 2 + 2P_m$

(b) $y(t) = [2 + 2m(t)] \cos(2\pi f_c t)$; Modulation index $\mu = \frac{m_p}{A} = \frac{2A_m}{2} = A_m$

(c) Output signal:

$$r(t) = 1 + m(t) + (1 + m(t)) \cos(4\pi f_c t)$$



(d) Modulation index $\mu \leq 1$ ensure non-coherent detection. $\frac{m_p}{A} \leq 1$; $\frac{2A_m}{2} \leq 1$ gives $A_m \leq 1$.

3. Solution

